

Monetary Policy

ECON 101 – Macroeconomics

Muhebullah Karimzada

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Chapter Outline

- **11.1** What Is Monetary Policy?
- **11.2** The Money Market and the Bank of Canada's Choice of Monetary Policy Targets
- **11.3** Monetary Policy and Economic Activity
- **11.5** A Closer Look at the Bank of Canada's Setting of Monetary Policy Targets

11.1 Learning Objective

Goal

Define monetary policy and describe the Bank of Canada's monetary policy goals.

- **Monetary policy:** actions the Bank of Canada (BoC) takes to manage the money supply and interest rates to pursue macroeconomic objectives.
- The BoC was established in **1934** with a mandate to act “in the best interest of the economic life of the nation.”
- Since WWII: the BoC conducts **active monetary policy** to stabilize inflation and output.
- **2022–2024 context:** the BoC conducted the most aggressive rate hiking cycle in 40 years to combat post-COVID inflation.

The Four Goals of Monetary Policy

Goal	Meaning and Canadian Context
Price Stability	Keep inflation low and stable. BoC's explicit target: 2% (within 1–3% band). Low inflation preserves the value of money and reduces uncertainty.
High Employment	Keep the economy near full employment. BoC monitors the unemployment rate, job vacancies, and wages as indicators of labour market slack.
Financial Stability	Ensure banks, credit markets, and financial infrastructure function smoothly. Key after 2007–09 global financial crisis.
Economic Growth	Provide a stable monetary environment that supports long-run real GDP growth.

The BoC's Dual Mandate in Practice

- The BoC's **primary operational goal** is the 2% inflation target (renewed with the federal government every 5 years — most recently 2021–26).
- But the BoC also cares about **employment and output**:
 - In 2021 renewal: the BoC added an explicit **labour market slack** consideration.
 - This moved Canada toward a **flexible average inflation targeting** (FAIT) framework, similar to the U.S. Fed.
- **Short-run trade-off**:
 - Reducing inflation too aggressively $\Rightarrow$ higher unemployment.
 - Easing to reduce unemployment $\Rightarrow$ risk of fuelling inflation.
- Monetary policy involves **difficult trade-offs**.

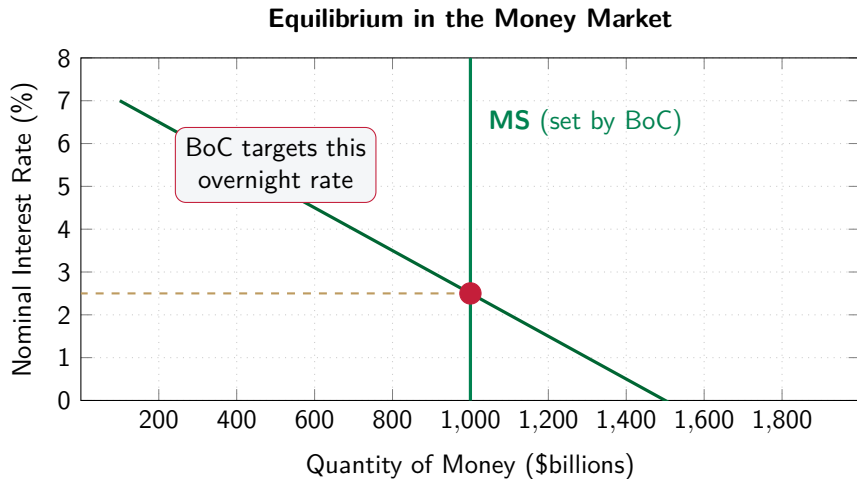
11.2 Learning Objective

Goal

Describe the Bank of Canada's monetary policy targets and explain how expansionary and contractionary policies affect interest rates.

- **BoC's two main tools:**
 - **Open market operations:** buying/selling Government of Canada bonds.
 - **Lending to financial institutions:** standing liquidity facilities.
- These affect **monetary policy targets:**
 - Money supply (secondary target).
 - Short-term overnight interest rate (**primary target**).

Figure 11.1 – The Money Market



Why Money Demand Slopes Downward

- **Why do people hold money?**
 - To make transactions (buy groceries, pay rent, tap an Interac machine).
- **Opportunity cost of holding money:** the interest rate forgone on bonds/savings.
- When interest rates are **high**:
 - Keeping cash in your wallet is costly — you're forgoing 5% interest on a GIC.
 - People shift funds from chequing accounts to savings accounts, bonds, GICs.
 - **Less money demanded.**
- When interest rates are **low**:
 - The opportunity cost of holding money is small.
 - People hold more money (liquid), less bonds.
 - **More money demanded.**

Shifts in Money Demand

- Money demand increases when:
 - Real GDP rises (more transactions needed).
 - Price level rises (each transaction requires more dollars).
- Money demand decreases when:
 - Real GDP falls (fewer transactions).
 - Technological improvements reduce need for cash (e.g., growth of e-transfers, tap payments).
- **Canadian example:** Canada has one of the highest contactless payment adoption rates globally — reducing the demand for physical cash (M1+ currency component has been flat while digital payments soar).

Figure 11.2 – Expansionary Monetary Policy (BoC Cuts Rate)

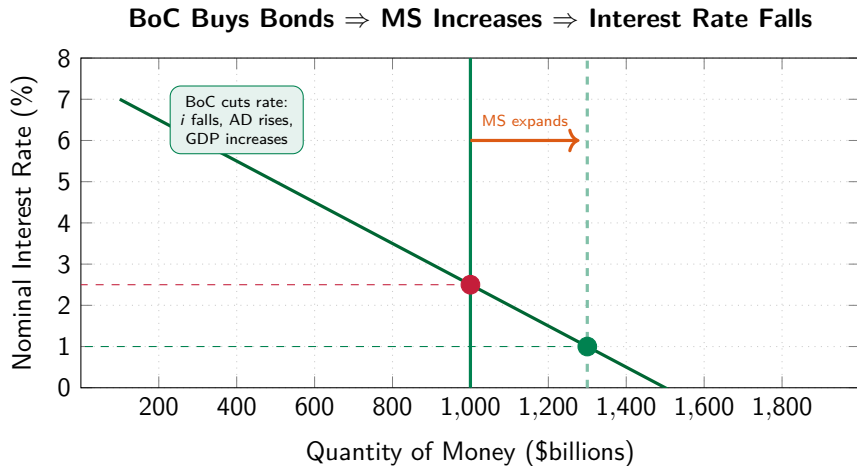


Figure 11.3 – Contractionary Monetary Policy (BoC Raises Rate)

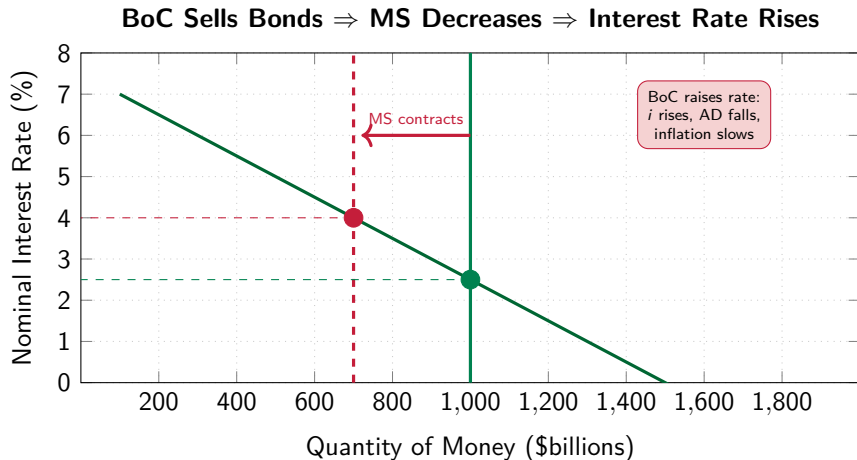
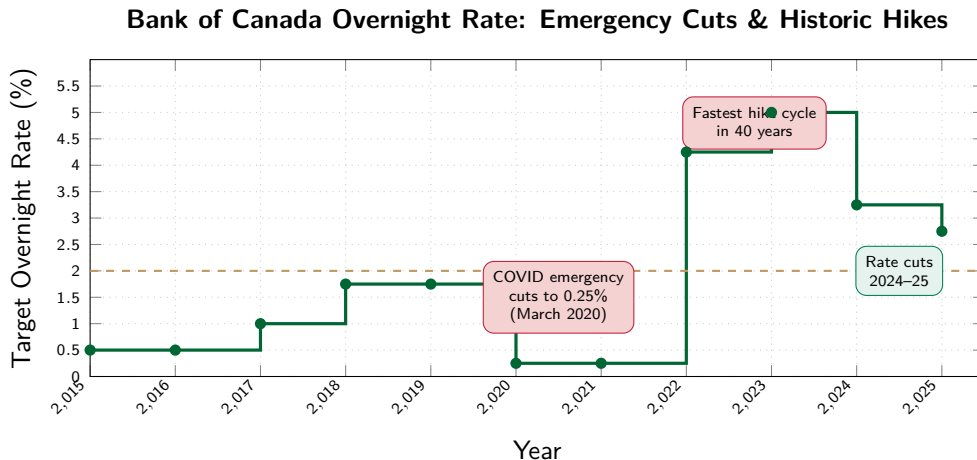


Figure 11.4 – Canada's Overnight Rate (2015–2025)



Source: Bank of Canada overnight rate press releases. 2025 value is estimated.

11.3 Learning Objective

Goal

Use AD-AS graphs to show the effects of monetary policy on real GDP and the price level.

- How does the overnight rate affect the real economy?
- **Transmission mechanism:**
 1. BoC changes overnight rate →
 2. Commercial banks adjust prime rate, mortgage rates, GIC rates →
 3. Households and firms change spending (C, I, NX) →
 4. AD shifts →
 5. Real GDP and inflation change.
- Typical lag: **12–18 months** from rate change to full inflation effect.

How Interest Rates Affect Aggregate Demand

1. Consumption (C)

- Lower rates $\Rightarrow$ cheaper consumer credit (car loans, home equity lines).
- **Key Canadian channel:** most mortgages renew every 5 years — rate hikes in 2022–23 raised mortgage payments by hundreds of dollars per month for millions of households, cutting disposable income and spending.

2. Investment (I)

- Lower rates $\Rightarrow$ cheaper to finance factories, equipment, housing.
- Residential construction is especially rate-sensitive.

3. Net Exports (NX)

- Higher Canadian rates attract foreign capital $\Rightarrow$ CAD appreciates.
- Stronger CAD makes Canadian exports more expensive $\Rightarrow$ NX falls.
- Weaker CAD (after rate cuts) makes Canadian goods cheaper for U.S. buyers.

Figure 11.5 – Expansionary Monetary Policy in AD-AS

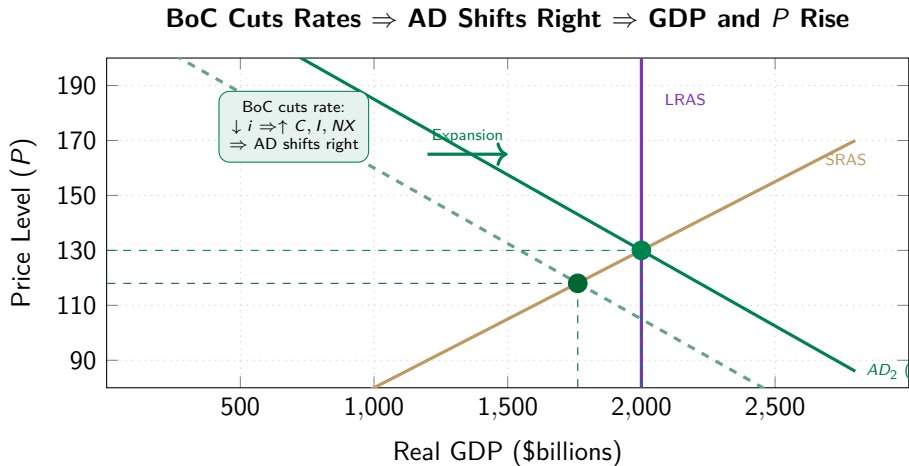
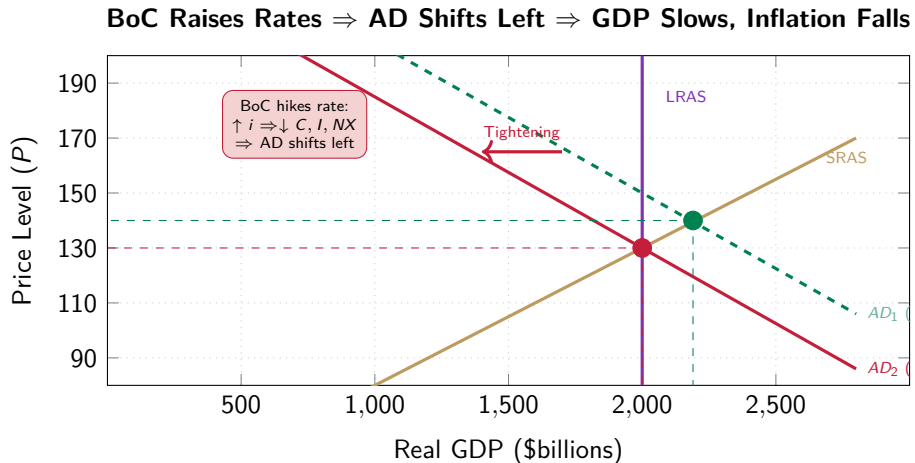


Figure 11.6 – Contractionary Monetary Policy in AD-AS



Can the BoC Eliminate Recessions?

- In textbook diagrams, the BoC:
 - knows exactly how far to shift AD, and
 - manages to shift it by precisely the right amount.
 - In reality, this is far harder:
1. **Data lag:** GDP data arrives with a 2–3 month delay. The BoC acts on **incomplete information**.
 2. **Decision lag:** the BoC meets 8 times per year to set the overnight rate. Policies take time to approve.
 3. **Effect lag:** interest rate changes take **12–18 months** to fully affect inflation and 6–12 months to affect real GDP.
- **Canada 2021–22 example:** BoC held rates at 0.25% too long while inflation was building $\Rightarrow$ was “behind the curve” $\Rightarrow$ had to hike very aggressively. The BoC acknowledged this in its 2023 review.

11.5 Learning Objective

Goal

Discuss the Bank of Canada's setting of monetary policy targets.

- Should the BoC target the money supply (M) or the interest rate (i)?
- What simple rules can guide monetary policy?
- What is **inflation targeting** and why did Canada adopt it?

Monetarism and the Money Growth Rule

- **Milton Friedman** (monetarists) advocated a **monetary growth rule**:
 - Increase the money supply at $\approx$ the long-run rate of real GDP growth ($\approx 2\%$).
 - This would produce stable prices with $\pi \approx 0$.
- Friedman argued:
 - Active monetary policy introduces unpredictability and may **destabilize** the economy.
 - A simple rule reduces discretionary error.
- **Why monetarism lost influence after the 1980s**:
 - The link between $M1+/M2$ growth and inflation became **unstable**.
 - Financial innovation changed the velocity of money unpredictably.
 - Modern economies have more complex transmission channels.

The BoC Can't Target Both M and i

- It may seem desirable to **simultaneously** target:
 - a specific money supply, and
 - a specific interest rate.
- But M and i are **linked through the money demand curve**:
 - At a fixed money demand, a given M implies a unique i .
 - You can choose M or i , but not both independently.
- **BoC's choice:** target the **overnight interest rate** (since 1996).
 - More transparent and more directly linked to economic conditions.
 - Communicates clearly to markets (rate announcements on fixed dates).

The Taylor Rule

- The **Taylor rule** (John Taylor, Stanford, 1993) provides a formula linking the overnight rate to economic conditions:

$$i_t = r^* + \pi_t + \frac{1}{2}(\pi_t - \pi^*) + \frac{1}{2}\left(\frac{Y_t - Y_t^*}{Y_t^*}\right),$$

where $r^* \approx 2\%$ (neutral real rate), $\pi^* = 2\%$ (target).

- **Interpretation:**

- If inflation $> 2\%$: raise i (fight inflation).
 - If inflation $< 2\%$: lower i (stimulate).
 - If $Y > Y^*$ (output gap positive): raise i .
 - If $Y < Y^*$ (output gap negative): lower i .
- The Taylor rule is a useful **benchmark**, not a rigid prescription.

Inflation Targeting: Canada's Framework

- **Canada adopted inflation targeting in 1991** — one of the first countries to do so.
- **Target:** CPI inflation of 2%, within a control band of 1–3%.
- **Why it works:**
 - Anchors **inflation expectations** — if people expect 2% inflation, they set wages and prices accordingly, which helps deliver 2% inflation.
 - Increases BoC **accountability** (must explain deviations publicly).
 - Provides clarity for households (mortgage planning) and firms (investment decisions).
- **Track record:** Canada averaged $\approx 2.0\%$ inflation 1991–2019 — remarkably close to target. The 2021–22 surge was the first major deviation in 30 years.
 - **Source:** Statistics Canada, CPI; Bank of Canada.

Summary

- **Monetary policy:** Bank of Canada actions to manage money supply and interest rates toward price stability, high employment, financial stability, and growth.
- BoC tools: open market operations and lending facilities $\Rightarrow$ affect the overnight rate.
- **Expansionary:** BoC buys bonds $\Rightarrow$ MS increases $\Rightarrow i$ falls $\Rightarrow C, I, NX$ rise $\Rightarrow$ AD shifts right $\Rightarrow$ GDP and P rise.
- **Contractionary:** BoC sells bonds $\Rightarrow$ MS decreases $\Rightarrow i$ rises $\Rightarrow C, I, NX$ fall $\Rightarrow$ AD shifts left $\Rightarrow$ inflation slows.
- Policy effectiveness limited by data, decision, and effect lags.
- The BoC **cannot simultaneously** target M and i — it chooses the overnight rate.
- The **Taylor rule** provides a useful benchmark for setting the overnight rate.
- Canada's **inflation targeting** (2% target since 1991) successfully anchored expectations for 30 years until the COVID-19 shock.
- **2022–24 case study:** BoC raised rate from 0.25% to 5.0% to tame 8.1% inflation; by late 2024, CPI returned to ≈ 1.8 – 2.7% , enabling rate cuts.

Thank you!

Questions?